Total No. of Pages: 2
VI SEMESTER
MID-SEM EXAMINATION

Roll No.....
B.TECH. [IT]
(March- 2019)

IT 306 ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS

Time: 1:30 Hours **Max. Marks : 20**

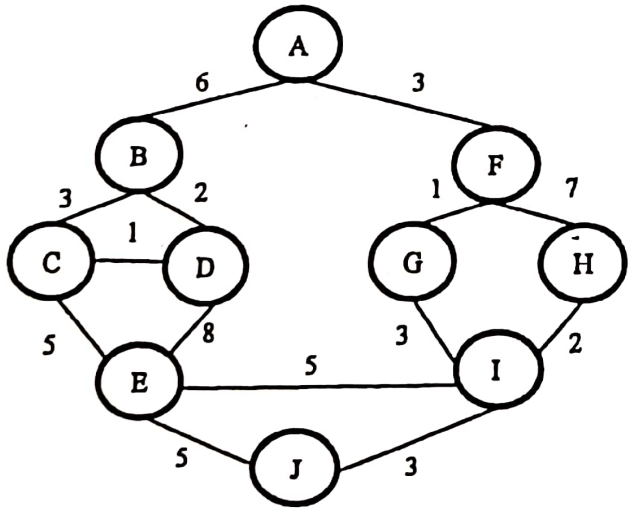
Note: Attempt All Questions.

Q1) Solve the following cryptographic problem by using a Generate and Test method, and applying the following constraints: (4)

a) Each letter represents exactly one number (in the range [0-9])
b) No two different letters represent the same number

```
  F O R T Y
    + T E N
    + T E N
 -----------
  S I X T Y
```

Q2) Define the term Heuristic. Find the shortest path from vertex A to vertex J using A* algorithm. (4)

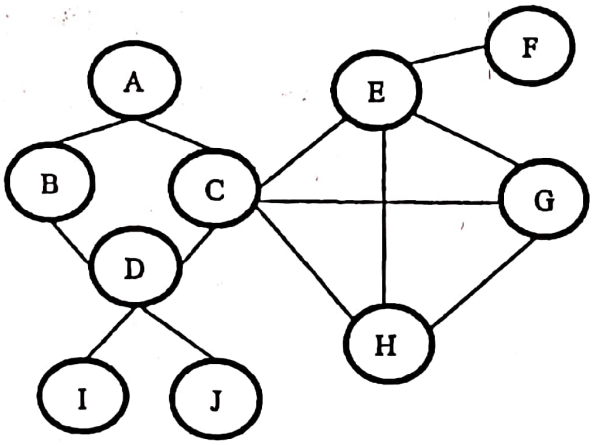


Node	Heuristic Value
A	10
B	8
C	5
D	7
E	3
F	6
G	5
H	3
I	1
J	0

P.T.O

Q3) We have two water jugs, one measures 4 Gallons (4G) while the other measure 9 Gallons (9G). But there is no measuring label mentioned on either of these two jugs i.e. we can't know the exact amount filled in the jug. Now, assuming there is an infinite amount of water supply, explain how we can measure 3G, 5G, 7G, and 8G using these unmarked jugs by representing it as a state space problem. (We can empty the jugs, and transfer water from one jug to other) (4)

Q4) Apply Depth First Search to find the traversal of the following graph by taking node "A" as start node and classify the edges after the traversal as tree edge, back edge, forward edge and cross edge. (4)



Q5) What are agents in Artificial Intelligence? Explain the structure of intelligent agents. (4)

-END-